

AKHIL RAJANA

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Objective

To start my professional career in the electronics and semiconductor industry by applying my knowledge of electronics, VLSI, circuit design, semiconductor devices, and programming to real-world engineering challenges. I aim to contribute to the organization while continuously learning new technologies and developing strong practical and technical skills.

Professional Summary

Electronics and Communication Engineering student with a foundation in analog electronics, CMOS circuit design, semiconductor devices, and VLSI. Hands-on experience with Cadence Virtuoso through analog IC design and circuit simulation projects, including a 45-nm CMOS dynamic comparator. Interested in applying circuit analysis, simulation, and semiconductor concepts to practical engineering problems.

Education

RNS Institute of Technology, Bengaluru, Karnataka	2023 – 2027
<i>B.E. – Electronics & Communication Engineering – CGPA: 8.77 / 10</i>	
Sri Venkateswara Junior College, Visakhapatnam, Andhra Pradesh	2022
<i>Intermediate – Percentage: 73.90 / 100</i>	
Dr K.K.R.'s Gowtham English Medium High School, Visakhapatnam, Andhra Pradesh	2020
<i>10th – Percentage: 74.40 / 100</i>	

Technical Skills

VLSI & IC Design: VLSI, Analog IC Design, CMOS Circuit Design, IC Layout, Semiconductor Devices.

EDA Tools: Cadence Virtuoso.

Programming: C, Embedded C, Verilog, Python.

Digital Electronics: Digital Logic Design, Sequential Logic, Timing Circuits.

Circuit Design: Circuit Simulation, Circuit Testing and Debugging, Hardware Debugging.

Projects

Implementation and Simulation of Power- and Energy-Efficient High-Gain Dynamic Comparator Using Stacking Floating Inverter Amplifier (FIA) Architecture in **Mar. 2026 – Oct. 2026**

45-nm CMOS | Ongoing

- **Mentor:** Ms. Dhanusha A
- **Team Size:** 4
- **Key Skills:** VLSI, Analog IC Design, 45-nm CMOS, Cadence Virtuoso, CMOS Circuit Design, Circuit Simulation, Digital Electronics.
- Working on the design and simulation of a high-gain dynamic comparator using the Stacking Floating Inverter Amplifier (FIA) architecture in 45-nm CMOS technology.
- Studying comparator architectures and implementing a dynamic pre-amplifier and regenerative latch to analyze gain, noise, power, and energy efficiency.

GSM-Based Traffic Signal Override for Emergency Vehicles

Aug. 2025 – Dec. 2025

- **Mentor:** Dr. Surah D
- **Team Size:** 4
- **Key Skills:** Arduino, GSM Modules, Embedded C, Real Time Systems, Circuit Design, Hardware Debugging.
- Developed a smart traffic control system using Arduino and GSM (SIM800L) to enable emergency vehicles to override traffic signals.
- Implemented secure code-based communication between transmitter and receiver units for signal control.
- Designed and integrated hardware components including keypad, microcontroller, and LED indicators.
- Contributed to circuit design, embedded C programming, and system testing.
- Achieved reduced delay for emergency vehicles and improved traffic management efficiency.

Traffic Light Circuit Using 555 Timer IC

Apr. 2024 – May 2024

- **Mentor:** Md. Ameen Sadiq S
- **Team Size:** 3
- **Key Skills:** Circuit Design, 555 Timer IC, Timing Circuits, Sequential Logic, Circuit Testing and Debugging.
- Designed and implemented a traffic light control system using 555 timer IC for sequential signal operation.
- Developed timing circuits using resistors and capacitors to control LED switching (Red, Yellow, Green).
- Built and tested the circuit on a breadboard to ensure accurate signal timing and reliability.
- Contributed to circuit design, component selection, and debugging.

Seminars / Trainings / Workshops

Bootcamp: RTL, IP Integration, and SoC Signoff

04 Feb. 2026 – 08 Feb. 2026

- **Institute:** National Institute of Electronics and Information Technology (NIELIT)
- **Key Skills:** RTL Design, Digital Logic Design, IP Integration, SoC Design, SoC Architecture Fundamentals, VLSI Fundamentals, Timing Analysis, Functional Verification, RTL Simulation, Synthesis.
- Completed a 30-hour bootcamp covering RTL design, IP integration, SoC design, and semiconductor design signoff, gaining practical exposure to the digital IC design flow.

Certifications

Bootcamp: RTL, IP Integration, and SoC Design Signoff

08 Feb. 2026

- **Issued by:** National Institute of Electronics and Information Technology (NIELIT)
- Certified in a 30-hour bootcamp covering RTL design, IP integration, SoC architecture, timing analysis, functional verification, RTL simulation, synthesis, and semiconductor design signoff.

Analog IC Design Certification

19 Aug. 2026

- **Issued by:** VTU-Centre for Online Education
- Covered advanced analog IC design and analysis, including CMOS op-amp design, negative feedback, stability, phase margin, circuit noise, mismatch, and compensated amplifier architectures.